SimLKAS: A Simulation-Based Framework for the Verification and Validation of Lane Keeping Assistance Systems

Tianjia Yang, Yuxin Ding, Ke Li, Yuyan Pan, Ruwen Qin, Zhaozheng Yin, Xianbiao Hu*

Tianjia Yang
Department of Civil and Environmental Engineering
Pennsylvania State University, University Park, PA, 16802
Email: tjyang@psu.edu

Yuxin Ding
Department of Civil and Environmental Engineering
Pennsylvania State University, University Park, PA, 16802
Email: ymd5170@psu.edu

Ke Li
Department of Civil Engineering
Stony Brook University, Stony Brook, NY 11794-4424
Email: ke.li.1@stonybrook.edu

Yuyan Pan
Department of Civil and Environmental Engineering
Pennsylvania State University, University Park, PA, 16802
Email: yypan@psu.edu

Ruwen Qin
Department of Civil Engineering
Stony Brook University, Stony Brook, NY 11794-4424
Email: ruwen.qin@stonybrook.edu

Zhaozheng Yin
Department of Computer Science
Stony Brook University, Stony Brook, NY 11794-4424
Email: zyin@cs.stonybrook.edu

Xianbiao Hu*
Department of Civil and Environmental Engineering
Pennsylvania State University, University Park, PA, 16802
Email: xbhu@psu.edu

*Corresponding author

Abstract: Lane Keeping Assistance Systems (LKAS) significantly enhance vehicle safety by maintaining lane centering through automated steering control. Evaluating LKAS performance in real-world conditions is challenging, as rare and extreme scenarios are difficult to capture systematically. Moreover, real-world testing is often time-consuming, costly, and lacks precise control over critical environmental factors, limiting its effectiveness for comprehensive assessment. However, existing simulation-based evaluations, such as those in CARLA, assume perfect pavement marking detection and neglect lateral deviations, limiting real-world applicability. To address these challenges, this study introduces SimLKAS, a simulation-based framework implemented in CARLA for systematic verification and validation of LKAS under diverse and challenging environments. The framework comprises three modules: context-based pavement marking perception, feedback-based loop controller, and ensemble learning-based performance evaluation. A key advantage of this modular framework is its analytical traceability for assessing the propagation of data inaccuracies and modeling errors across layers, along with its algorithmic flexibility for integrating alternative algorithms at each stage. Extensive experiments confirm that SimLKAS generates diverse testing conditions while reproducing realistic lane-keeping patterns consistent with previous studies, validating its effectiveness for LKAS assessment. The results, analyzed using Random Forest regression and SHAP analysis, indicate that heavy rain, high speed, and curved roads significantly contribute to increased lateral deviation in daytime conditions, whereas nighttime performance is strongly influenced by streetlights and vehicle headlight settings. This framework provides researchers and practitioners with a scalable, repeatable, and controlled platform for benchmarking LKAS algorithms, optimizing performance, and enhancing safety validation, thereby supporting advancements in autonomous driving technology.

Keywords: Automated Vehicles, Lane Keeping Assistance System, Verification and Validation (V&V), Lane Detection, Vehicle Lateral Control.

1. INTRODUCTION

Automated vehicle (AV) technologies have evolved from conceptual research to real-world implementation, with many modern vehicles now incorporating advanced driver-assistance systems (ADAS). Among these, Lane Keeping Assistance Systems (LKAS) are particularly critical for enhancing road safety by preventing unintentional lane departures. LKAS relies on vision-based sensors, such as cameras, combined with advanced image processing algorithms to detect lane markings and monitor the vehicle's lateral position. By providing automated steering control, LKAS helps maintain lane centering and improves driving stability. A 2018 study estimated that up to 27% of head-on and single-vehicle crashes could be prevented if LKAS were widely deployed in Finland (Utriainen et al., 2020). Recognizing these safety benefits, LKAS adoption has surged in recent years, with over 86% of new vehicles in the U.S. equipped with LKAS as of 2023 (MITRE, 2024). As LKAS becomes increasingly prevalent, ensuring its safety, reliability, and effectiveness under diverse real-world conditions has become a key research focus.

To ensure safety, automation manufacturers are required to conduct verification and validation (V&V) testing of LKAS in compliance with industry standards such as ISO 11270. However, these standards establish only minimal performance requirements and are typically tested under controlled conditions, such as straight or curved roads with clear lane markings on closed test tracks (Wei et al., 2023). They do not comprehensively evaluate LKAS performance in complex real-world scenarios, particularly under adverse weather conditions where visibility is reduced, or lane markings become less distinguishable. Recognizing these limitations, LKAS vendors include disclaimers in their manuals, stating that the system may not function reliably in poor weather and reminding drivers to remain in control at all times. However, drivers may occasionally overestimate the system's capabilities, which can be particularly dangerous in extreme conditions where degraded LKAS performance may fail to provide adequate support, increasing the risk of lane departure if drivers are not actively engaged.

Moreover, field testing depends on naturally occurring weather conditions, making it time-consuming, costly, and potentially unsafe. More critically, real-world studies struggle to capture rare but safety-critical scenarios, such as specific combinations of rain, lighting, speed, and road curvature, which do not occur frequently enough for systematic evaluation. Consequently, existing research provides limited insights into LKAS performance under corner cases, where the system's reliability is most crucial. Additionally, many studies focus on binary outcomes—such as whether LKAS successfully detects lane markings—rather than conducting a detailed quantitative analysis of lateral deviations to assess how well the system maintains lane centering. A more granular evaluation of vehicle lateral control is essential for identifying performance degradation factors and guiding future improvements. Researchers have also examined how lane marking characteristics—such as width, color, contrast, continuity, and retroreflectivity—affect LKAS performance (Formosa et al., 2024; Pike et al., 2024; Shaon et al., 2024), offering valuable insights for pavement marking enhancements. However, these studies do not comprehensively assess LKAS performance across diverse environmental conditions. Neumeister and Pape (2019) conducted field tests to explore this issue, but real-world testing remains logistically challenging, costly, and inherently constrained. These limitations highlight the urgent need for a systematic and controlled evaluation framework that enables LKAS performance assessment under diverse and extreme conditions.

Simulation-based testing provides a cost-effective, scalable, and repeatable alternative to real-world LKAS evaluation. However, most simulators, such as CARLA and SUMO, assume perfect pavement marking detection and ideal vehicle trajectory control, which do not reflect real-world driving conditions. Specifically, lane detection in CARLA is assumed to be flawless, and vehicles exhibit near-perfect lane centering with minimal lateral deviation, failing to account for perception errors and natural trajectory variations. This simplification limits the applicability of simulation results for evaluating LKAS performance in realistic scenarios. Comparative studies of SUMO, CARLA, and real-world trajectories (Qi, 2024; Qi & Hu, 2023) reveal that while vehicles in SUMO and CARLA remain nearly centered within the lane with little fluctuation, real-world vehicles naturally exhibit continuous lateral variations due to factors such as sensor noise, steering corrections, and road surface irregularities. These discrepancies underscore

the need for a more realistic simulation framework that accounts for imperfect lane detection, lateral deviations, and real-world driving uncertainties to enable a more accurate and comprehensive assessment of LKAS performance under diverse environmental conditions.

To address the limitations of existing LKAS V&V testing, this research introduces a new modular and controlled simulation-based framework that enables precise manipulation of key environmental and operational factors. Unlike real-world testing, which struggles to capture rare but safety-critical scenarios, this framework allows for the systematic generation of diverse test conditions that may not frequently occur in practice. The controlled variables include road geometry, vehicle speed, weather conditions, lighting settings and others. To achieve this, the framework comprises three core modules: context-aware pavement marking perception, feedback-based loop controller, and ensemble learning-driven performance evaluation. The perception module employs vision-based lane detection algorithms, integrating edge detection and clustering techniques to extract lane markings and determine the ego vehicle's lateral position relative to the detected lane boundaries. The control module implements a feedback-based loop controller, continuously monitoring the state of the LKAS system and generate precise steering commands based on the difference between the desired setpoint and the actual output, ensuring robust lane centering. The evaluation module applies an ensemble learning framework, incorporating Random Forest regression and SHAP (Shapley Additive Explanations) analysis, to systematically assess LKAS performance by quantifying lateral deviations under varying environmental and operational conditions. By leveraging a high-fidelity simulation environment, this framework enables structured, repeatable, and scalable LKAS evaluations, ensuring rigorous assessment under diverse and challenging scenarios.

A key advantage of this framework is its modular structure, allowing for analytical traceability and algorithmic flexibility. Each module is explicitly defined and analytically derivable, providing a clear understanding of data flow and error propagation across different layers. This feature enables researchers to evaluate how data inaccuracies and modeling errors at the perception level influence control decisions and overall system performance. Additionally, the modular design allows for seamless integration of alternative algorithms, making it adaptable for future advancements in LKAS technologies. By offering a controllable and structured approach to performance evaluation, this framework serves as a valuable tool for benchmarking LKAS algorithms and refining lane-keeping strategies.

To validate the framework, a comprehensive set of test scenarios was generated, systematically varying road geometry, vehicle speed, weather conditions, and lighting settings, demonstrating the framework's ability to create diverse and challenging conditions for LKAS V&V. Additionally, a comparative analysis of vehicle behavior patterns with findings from field data and prior studies confirmed that the proposed framework accurately reproduces realistic driving behaviors, further validating its effectiveness for LKAS assessment. The evaluation employed Random Forest regression with SHAP analysis to identify key factors influencing LKAS performance. The results indicate that heavy rain, high speed, and curved roads significantly increase lateral deviation during daytime conditions, whereas nighttime performance is primarily affected by streetlight availability and vehicle headlight settings. These findings demonstrate the framework's capability to generate diverse testing conditions and systematically assess LKAS performance degradation.

To summarize, the proposed structured simulation framework provides a scalable, repeatable, and controlled testing environment, supporting the development, benchmarking, and refinement of LKAS technologies while enhancing the safety and reliability of autonomous driving systems. The key contributions of this study are as follows:

1. Development of a modular and controlled simulation-based framework, integrating context-aware pavement marking perception, feedback-driven lateral control, and ensemble learning-based performance evaluation. This architecture allows for analytical traceability, modular algorithm replacements, and detailed analysis of data flow and error propagation, ensuring adaptability for future research.

2. Systematic generation of diverse testing scenarios, enabling a comprehensive and structured evaluation of LKAS performance across varying settings. Additionally, a comparative analysis of vehicle behavior confirmed that the proposed framework accurately replicates real driving patterns, validating its effectiveness for LKAS assessment.
3. Identification of key factors contributing to LKAS performance degradation, demonstrating the influence of weather conditions, vehicle speed, and road curvature on lateral deviation through data-driven analysis using Random Forest regression and SHAP interpretability techniques.

The remainder of the paper is organized as follows: Section 2 provides a comprehensive review of relevant literature, covering lane detection techniques, vehicle lateral control methods, and existing LKAS evaluation research. Section 3 details the proposed framework and the methods employed in each module. Section 4 describes the experimental settings. Section 5 presents the LKAS performance evaluation and discusses potential improvements. Finally, Section 6 summarizes the key findings and offers suggestions for future work.

2. LITERATURE REVIEW

Since the proposed evaluation framework consists of three modules, we review the relevant literature in these three key areas: lane detection, vehicle lateral control, and LKAS evaluation.

2.1 Research on Lane Detection

Lane detection is a critical application of computer vision, a field that has advanced significantly in recent years. Several comprehensive reviews of lane detection methodologies have been conducted (Tang et al., 2021; Zakaria et al., 2023). Lane detection algorithms can be broadly categorized into traditional image-processing techniques and deep learning-based methods.

Traditional approaches typically follow a three-step process: image preprocessing, feature extraction, and lane model fitting (Assidiq et al., 2008). Image preprocessing enhances image quality by removing noise, with a common strategy being the definition of a Region of Interest (ROI). Since onboard cameras are typically mounted near the top of the vehicle and face forward, the upper portion of the captured image—containing irrelevant background elements such as the sky and buildings—is cropped out ((Ding et al., 2013; Hsiao et al., 2008). Fixed-size ROIs remain the most widely used approach. Additionally, images are often converted to grayscale to reduce redundant color information. Feature extraction techniques help identify lane pixels from preprocessed images. Common methods include filtering techniques, edge detection algorithms, and clustering methods (Aung & Zaw, 2014; Ghanem et al., 2023; Mu & Ma, 2014; Wu et al., 2014). Finally, lane model fitting applies mathematical models to represent detected lane markings using approaches such as Hough Transform, Random Sample Consensus (RANSAC), and cubic B-spline fitting (Hsiao et al., 2008; Kim, 2008; Niu et al., 2016).

With the rise of machine learning, deep learning-based approaches have emerged as powerful tools for lane detection. Convolutional Neural Networks (CNNs) are widely used due to their strong pattern recognition capabilities and have been extended with modules such as Long Short-Term Memory (LSTM) to capture temporal dependencies between frames (Kim & Lee, 2014; Zou et al., 2019). To overcome the limitations of earlier models, including reliance on extensive preprocessing and the inability to detect a flexible number of lanes, end-to-end architectures like LaneNet have been proposed (Neven et al., 2018). More recent advancements incorporate generative adversarial networks (GANs), knowledge distillation, and transformers (Ghafoorian et al., 2018; Hou et al., 2019; Zhang et al., 2021). Despite their effectiveness, deep learning-based lane detection methods face generalization challenges and require large, high-quality labeled datasets for training, limiting their practical deployment.

2.2 Vehicle Lateral Control Models

Vehicle control consists of two primary components: longitudinal and lateral maneuvers. Longitudinal maneuvers regulate throttle and braking to control acceleration and deceleration, while lateral maneuvers

adjust the steering angle to maintain lane centering. This review focuses on lateral control strategies used in LKAS, which can be categorized into feedback control methods and learning-based control methods (Liu et al., 2023).

Feedback control methods, such as Proportional-Integral-Derivative (PID) control and Model Predictive Control (MPC), rely on real-time vehicle dynamics and environmental data for decision-making. PID controllers are widely used in the industry due to their simplicity, ease of implementation, and robust performance, with various adaptations developed to address specific control challenges (Marino et al., 2011; Zhao et al., 2012). MPC is another widely used approach for vehicle control. It predicts the vehicle's future trajectory over a finite time horizon and computes optimal steering commands to minimize deviation from the desired path. MPC continuously updates its predictions using real-time feedback while incorporating vehicle dynamics constraints, resulting in smooth and stable lane tracking (Cheng et al., 2020; Chu et al., 2022; Liu et al., 2025). However, the optimization process introduces significant computational complexity, posing challenges for real-time decision-making and practical field implementation.

In recent years, learning-based control methods have gained attention due to their ability to address nonlinear control problems without relying on explicit vehicle models. Among them, Deep Reinforcement Learning (DRL) has emerged as a promising approach, with numerous studies investigating its application in vehicle lateral control (Li et al., 2019; Pérez-Gil et al., 2022; Wang et al., 2023; Wang et al., 2019). In addition to DRL-based low-level controllers that generate steering commands based on upper-level planning modules, end-to-end learning frameworks have also been explored. These models take raw sensor inputs, such as camera images, and directly output control commands without explicit feature extraction or planning stages (Amini et al., 2020; Chen & Huang, 2017; Sallab et al., 2016; Wang et al., 2022). However, although learning-based control exhibits promising performance, it inherently has several limitations. For example, learning-based approaches require large amounts of data for training, and most existing research relies on simulation environments for model development. The transition from simulation to real-world implementation remains a significant challenge (Chen et al., 2024). Additionally, safety is a critical concern, as DRL often lacks interpretability, making it difficult to fully understand the decision logic of the agent, which poses challenges for deploying such systems (Kuutti et al., 2021).

2.3 LKAS Evaluation Standards and Studies

In recent years, LKAS adoption has increased rapidly. As shown in Figure 1, the share of new vehicles in the U.S. equipped with LKAS has grown from only 1.4% in 2015 to 86.3% in 2023, a figure that continues to rise (MITRE, 2024).

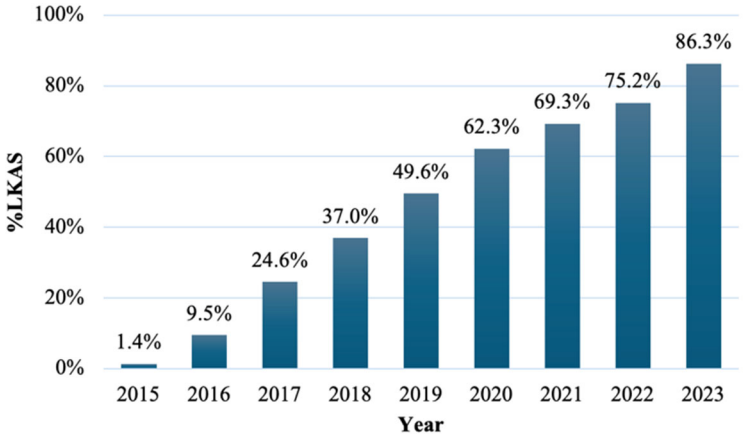


Figure 1. Growth of LKAS Adoption in the U.S.

With this widespread adoption, several standards have been introduced to establish performance benchmarks. These includes ISO-17361, ISO-11270, EURO NCAP, and UN ECE R79, which define minimal requirements for LKAS functionality (Wei et al., 2023). These standards primarily specify

controlled test scenarios and passing criteria. For example, ISO 11270 mandates testing on straight and curved roads at constant speeds between 20 and 22 m/s, with passing thresholds for lateral deviation set at 0.4 m for light vehicles and 1.1 m for heavy vehicles. National Highway Traffic Safety Administration (2024) also provides standardized LKAS testing procedures, requiring tests under ideal weather conditions with high-quality pavement markings conforming to the Manual on Uniform Traffic Control Devices (MUTCD). These standards focus on assessing LKAS in standardized conditions rather than evaluating its performance under diverse real-world scenarios.

Recent studies have evaluated the performance of LKAS beyond standardized tests, with a primary focus on the visibility requirements of pavement markings (Burghardt et al., 2021; Formosa et al., 2024; Hadi & Sinha, 2011; Marr et al., 2020; Pike et al., 2024; Pike et al., 2018; Shaon et al., 2024). These studies assessed factors such as lane marking width, color, contrast, continuity, and retroreflectivity in various test environments, offering practical recommendations for MUTCD pavement marking standards. Some researchers have also explored additional factors, such as lighting, glare, and adverse weather (Neumeister & Pape, 2019). However, these real-world studies often rely on simple evaluation metrics, primarily determining whether the LKAS can detect the left, right, or both lane markings. Consequently, their analysis lacks detailed assessments to quantify the impact of various factors on overall LKAS performance. Wang et al. (2025) addressed this by collecting detailed Controller Area Network (CAN) data and environmental context from over 7,000 km of driving. Still, field tests inherently have several limitations, including being time-consuming and financially intensive. Additionally, external factors, particularly weather conditions, remain uncontrollable. Moreover, conducting field tests in adverse conditions may pose significant safety risks.

In summary, lane detection methods have evolved from traditional image-processing techniques to deep learning-based approaches, with the latter demonstrating superior performance in complex environments. However, deep learning models face generalization challenges and require extensive labeled datasets, limiting their practical deployment. For vehicle lateral control, widely used feedback-based methods such as PID controllers offer simplicity and robustness, while optimization-based approaches like MPC provide precise control but at the cost of high computational complexity. More recently, DRL-based control strategies have gained attention due to their adaptability to nonlinear driving environments, but their reliance on large-scale training data and lack of interpretability pose challenges for real-world implementation. In terms of LKAS evaluation, most existing studies rely on real-world testing, which is costly, time-consuming, and inherently constrained by environmental conditions, limiting the ability to systematically assess performance across diverse scenarios. While some field studies have attempted to evaluate LKAS under adverse weather conditions, the inability to isolate specific factors and the risks associated with extreme testing highlight the need for a structured, scalable, and controlled simulation-based evaluation framework to comprehensively analyze LKAS performance across a wide range of operating conditions.

3. DEVELOPMENT OF SIMLKAS FRAMEWORK AND KEY TECHNICAL MODULES

The proposed framework, illustrated in Figure 2, comprises three main modules: context-aware pavement marking perception, feedback-based loop controller, and ensemble learning-driven performance evaluation. It is designed to simulate and analyze LKAS performance under a wide range of driving conditions. The framework is implemented in the CARLA simulation environment, which provides high-fidelity vehicle dynamics, realistic sensor models, and highly customizable environmental settings, ensuring controlled and repeatable experiments. By enabling precise manipulation of key parameters, including weather conditions, road geometry, vehicle speed, streetlight availability, and vehicle lighting settings, the framework facilitates a systematic and structured evaluation of LKAS performance under diverse and challenging scenarios. While the core techniques are grounded in established literature, the framework incorporates tailored integration and engineering refinements, such as perception error mitigation in the control feedback loop, to ensure stable performance under clear weather conditions.

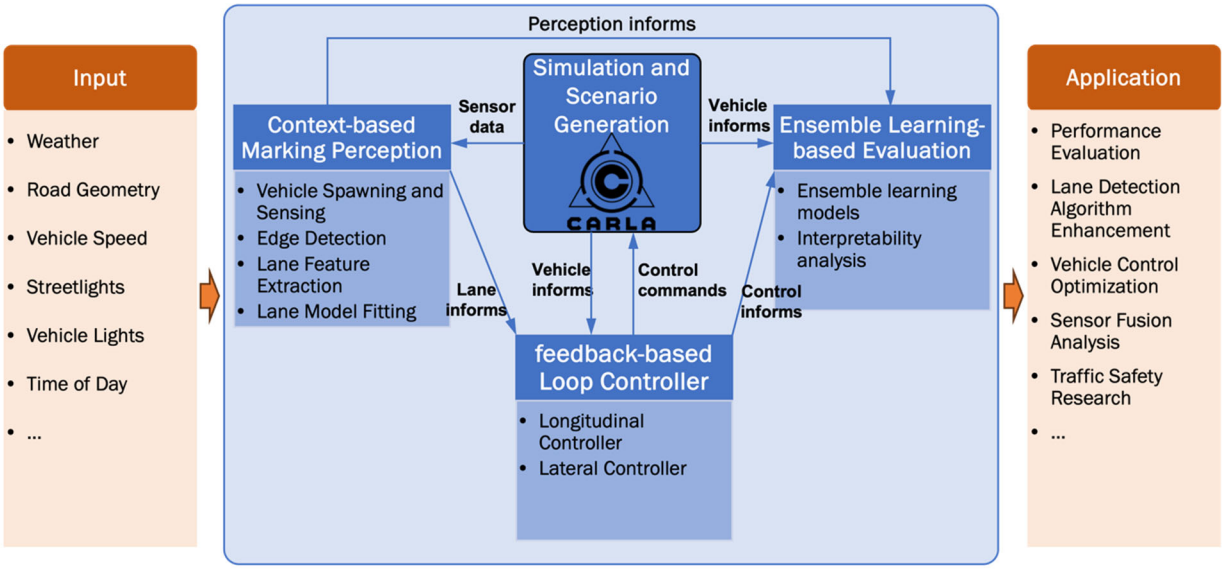


Figure 2. Illustration of SimLKAS Framework and Key Modules

The context-aware pavement marking perception module processes sensor data, primarily from onboard cameras, to detect lane boundaries and estimate the vehicle's lateral position within the lane. It employs edge detection, lane feature extraction, and lane model fitting techniques to interpret lane markings and derive mathematical representations of lane geometry. These processed lane features serve as inputs for downstream control algorithms, enhancing lane-keeping accuracy and robustness.

The feedback-based loop controller consists of longitudinal and lateral controllers, responsible for generating real-time control commands for steering, throttle, and braking. The longitudinal controller adjusts throttle and braking to maintain the target speed, while the lateral controller—the primary focus of this research—generates steering commands to keep the vehicle centered within the lane. The framework processes sensor inputs at 60 frames per second, enabling continuous and adaptive lane-keeping adjustments that allow the ego vehicle to respond effectively to dynamic driving conditions.

The goal of the evaluation module is to analyze the influence of each factor to support further improvement. It collects and stores vehicle behavior data, perception accuracy metrics, and control responses throughout the simulation. After data collection, analysis techniques are applied to extract meaningful insights. In this study, traditional statistical regression methods were initially considered due to their interpretability, but they were found inadequate for capturing the nonlinear and interactive nature of the data. Therefore, an ensemble learning method, Random Forest regression—was selected for its ability to model complex relationships and its built-in compatibility with SHAP analysis for assessing feature importance. This approach enables robust and interpretable quantification of the impact of various external factors on LKAS effectiveness.

The modular architecture of the framework supports the integration of alternative lane detection algorithms, control strategies, and evaluation techniques, making it adaptable for a wide range of applications. Beyond LKAS performance assessment, potential applications include lane detection algorithm enhancement, vehicle control optimization, sensor fusion analysis, and traffic safety research. The following sections provide a detailed description of the methodologies implemented in each module within this study.

3.1 Context-based Pavement Marking Perception Module

The perception module is designed to detect lane markings across a wide range of driving conditions, including challenging environments where visibility may be reduced. The input for lane detection consists of video frames captured from the vehicle's onboard camera. In this research, we employ a systematic image processing approach, incorporating image preprocessing and edge detection, lane feature extraction,

and lane model fitting. This extracted data is then fed into the control module, enabling real-time lane-keeping adjustments. The complete processing pipeline is illustrated in Figure 3.

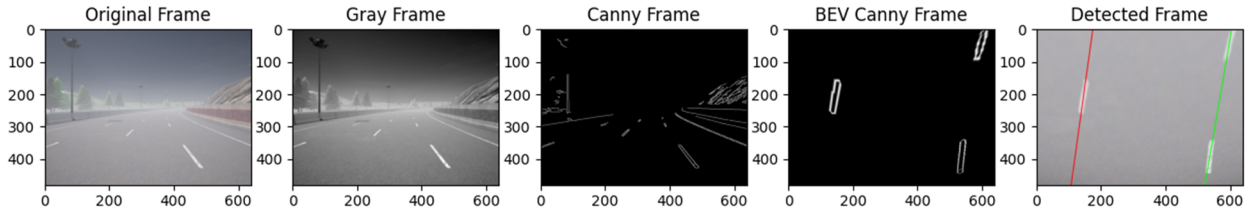


Figure 3. Illustration of Lane Detection Pipeline

3.1.1 Preprocessing and Edge Detection

To enhance the visibility of structural details such as lane markings, we first convert the input image from an RGB color space to grayscale. This transformation is defined mathematically as Eq. (1):

$$I(x,y) = 0.299R(x,y) + 0.587G(x,y) + 0.114B(x,y) \quad (1)$$

where $I(x,y)$ is the grayscale intensity at pixel (x,y), and R, G, B represent the red, green, and blue channel intensities, respectively. The weights in this formula originate from the luminance perception model used in the ITU-R BT.601 standard, which accounts for the varying sensitivity to different colors—being most sensitive to green, followed by red, and least sensitive to blue. This weighted sum operation is applied pixel-wise across the entire image, producing a grayscale image where higher intensity values correspond to brighter regions, enhancing contrast between lane markings and the surrounding road surface. This transformation simplifies subsequent edge detection and feature extraction tasks, as illustrated in the Gray Frame of Figure 3.

Next, we adopt the Canny Edge Detection algorithm to identify edges in the grayscale image Canny (1986). This algorithm consists of multiple stages designed to enhance edge detection robustness by eliminating irrelevant details and preserving key structural features, such as lane markings. To reduce noise and improve edge detection accuracy, a Gaussian filter is first applied to the image. Since raw images often contain noise that can lead to false edge detection, Gaussian smoothing helps suppress small variations while preserving dominant edges. The Gaussian filter, denoted as $G_\sigma(x,y)$, assigns greater weight to central pixels, thereby smoothing abrupt intensity changes and removing fine details that do not contribute to lane detection. The smoothed image, $I_s(x,y)$, is obtained as follows:

$$I_s(x,y), = G_\sigma(x,y) * I(x,y) \quad (2)$$

Where $G_\sigma(x,y)$is defined as:

$$G_\sigma(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (3)$$

Here, σ controls the degree of smoothing, and a larger σ results in stronger noise reduction at the cost of edge sharpness.

Edges in an image correspond to regions of rapid intensity change. Therefore, the next step is to compute the image intensity gradients in both the horizontal (G_x) and vertical (G_y) directions.

Instead of computing the first derivative directly, the Sobel filter is employed to approximate intensity gradients in both directions, as it is computationally efficient and well-suited for real-time applications. The gradient components G_x and G_y are computed as:

$$G_x = I_s(x,y) * S_x, \text{ where } S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad (4)$$

$$G_y = I_s(x,y) * S_y, \text{ where } S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad (5)$$

By combining G_x and G_y, the gradient magnitude, which determines the edge strength, is computed as:

$$G = \sqrt{G_x^2 + G_y^2} \quad (6)$$

The gradient direction is given by:

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right) \quad (7)$$

After computing the gradients, detected edges may appear thick and imprecise. To refine the edge map, non-maximum suppression is applied by retaining only local maxima along the gradient direction θ. Given a pixel (x, y), its two nearest neighboring pixels in the gradient direction are examined.

If its gradient magnitude is not the highest among its local neighbors, it is suppressed:

$$I_{thin}(x,y) = \begin{cases} G(x,y), if\ G(x,y) > G(x_n,y_n)\ and\ G(x,y) > G(x_m,y_m) \\ 0, otherwise \end{cases} \quad (8)$$

where (x_n, y_n) and (x_m, y_m) represent the two nearest neighboring pixels along the gradient direction θ, with (x_n, y_n) located in the forward direction of the gradient and (x_m, y_m) in the opposite direction.

To further refine the edge map and eliminate weak or noisy detections, a dual-thresholding approach is applied using an upper threshold T_h and a lower threshold T_l:

$$I_{edge}(x,y) = \begin{cases} G(x,y), if\ G(x,y) > T_h\ (strong\ edge) \\ G(x,y), if\ T_l < G(x,y) \le T_h\ and\ connected\ to\ a\ strong\ edge \\ 0, otherwise \end{cases} \quad (9)$$

This ensures that weak edges disconnected from strong edges are suppressed, preventing noise-induced false detections while preserving lane markings and critical structural details.

Since the input image is captured from a front-facing camera, lane markings appear distorted due to perspective effects. Distant lane markings converge, making it difficult to accurately measure lane width and curvature. To correct this, a homographic transformation is applied to warp the image into a bird's-eye (top-down) view, eliminating perspective distortion and ensuring consistent lane geometry representation.

A homography transformation is a projective mapping between two planes and is mathematically defined as:

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (10)$$

where (x, y) are original pixel coordinates, (x', y') are transformed coordinates in the bird's-eye view, w' is the homogeneous coordinate scaling factor, and H is a 3×3 perspective transformation matrix.

The transformation matrix H is computed using four reference points from the original image and their corresponding positions in the bird's-eye view. Given a set of four known correspondences (x_i, y_i) in the original image and their mapped positions (x_i', y_i') in the top-down view, H is determined by solving:

$$AH = 0 \quad (11)$$

where A is a coefficient matrix derived from the known point correspondences. Once H is determined, the bird's-eye view image is generated by applying the transformation to each pixel in the original image using Eq. 10.

Since the transformed pixel coordinates (x', y') may not align exactly with integer pixel locations, bilinear interpolation is used to compute the final pixel intensities, ensuring subpixel precision and smooth transformations. To maintain image quality, inverse warping is applied, mapping each pixel in the transformed image back to its corresponding location in the original image, thereby preserving structural consistency and minimizing artifacts.

After applying the Bird's-Eye View (BEV) transformation, a Region of Interest (ROI) is defined to exclude irrelevant portions of the image, such as the sky, distant scenery, and adjacent lanes that do not contribute to lane detection. Because the camera is front-facing and mounted near the top of the vehicle, lane markings predominantly appear in the lower portion of the BEV image. By leveraging the fixed spatial relationship between the camera and the lane boundaries, we employ a fixed ROI to focus detection on the immediate left and right lanes. This approach improves computational efficiency and reduces false detections. While effective in structured road environments, the fixed ROI method has limitations in complex or highly curved scenarios. To address these challenges, we plan to explore adaptive ROI strategies or multi-sensor fusion in future work.

The ROI is defined using a binary mask function $M(x, y)$, which retains pixels in the relevant road region and discards the rest, formulated as:

$$M(x,y) = \begin{cases} 1, y > H_{cutoff} \\ 0, y \leq H_{cutoff} \end{cases} \tag{12}$$

where H_{cutoff} is a fixed threshold determined based on the camera position and orientation. Applying this mask to the bird's-eye view image $I_B(x, y)$ produces the ROI-processed image $I_{ROI}(x, y)$:

$$I_{ROI}(x,y) = I_B(x,y) \cdot M(x,y) \tag{13}$$

3.1.2 Lane Point Clustering

Once edges are extracted, a clustering approach is applied to group lane pixels into meaningful lane structures, rather than directly fitting a line to all detected edges. In this research, we employ Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to separate lane markings from noise and irrelevant edges. Unlike conventional clustering algorithms such as K-Means, which assume compact and spherical clusters, DBSCAN is better suited for detecting elongated structures like lane markings. Additionally, DBSCAN does not require a predefined number of clusters, making it adaptable to varying road conditions where the number and appearance of lane markings may change dynamically.

DBSCAN operates based on two key parameters:

- ε: The maximum distance within which points are considered neighbors.
- Minimum Points (MP): The minimum number of points required to form a valid cluster.

In our context, the Euclidean distance between two lane pixel points (x_i, y_i) and (x_j, y_j) is given by:

$$d_{ij} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \tag{14}$$

A point p is classified as a core point if it satisfies the density condition:

$$|N_\varepsilon(p)| \geq MP \tag{15}$$

where $N_\varepsilon(p)$ represents the set of all points within a radius ε of p:

$$N_\varepsilon(p) = \{q \in D \mid d(p,q) \leq \varepsilon\} \tag{16}$$

where D is the entire dataset of lane edge points.

- If a point is reachable from a core point but does not satisfy the MP requirement, it is classified as a border point.

- Points that are neither core points nor border points are treated as noise and are excluded from further analysis.

The clustering process begins by randomly selecting a point. If the point is classified as a core point, a new cluster is initialized. The algorithm then expands this cluster by iteratively including neighboring points within ε, recursively growing the cluster as additional neighbors are found. This process is illustrated in Figure 4, which presents a real-world example from our detection pipeline. This process is illustrated in Figure 4, which presents a real-world example from our detection pipeline. The figure shows that lane point clustering starts from the BEV Canny image (introduced in Figure 3). To better illustrate noise filtering, synthetic noise has been added to the input. The resulting DBSCAN output displays each cluster in a distinct color, with noise points marked in gray.

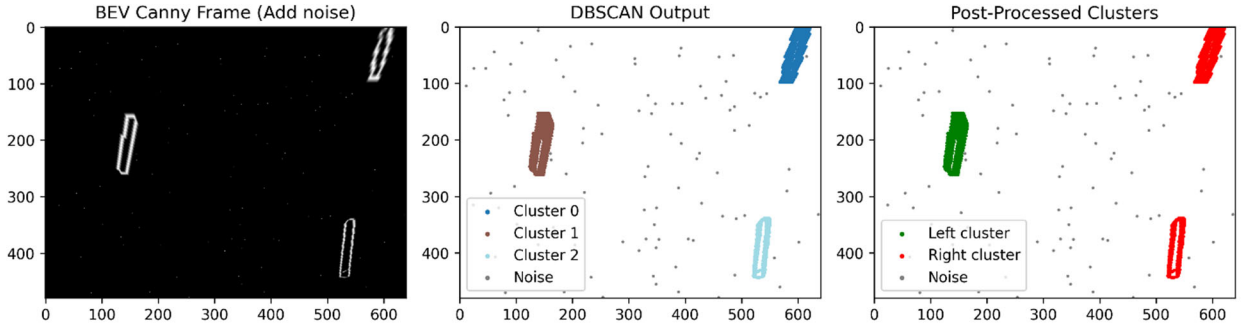


Figure 4. Illustration of the Lane Point Clustering process

After clustering, each detected lane marking is expected to form a distinct cluster. The extracted clusters are then categorized into left and right lane markings based on their mean x-coordinates in the image space. Given that the vehicle's position is centered in the image, lane classification is performed as follows:

$$C_k^{lane} = \begin{cases} C_k^{left}, if\ \overline{x}_k < x_{center}\ (left\ lane) \\ C_k^{right}, if\ \overline{x}_k \geq x_{center}\ (right\ lane) \end{cases} \quad (17)$$

where C_k^{left} and C_k^{right} represent the corresponding detected lane clusters, $\overline{x}_k$ is the mean x-coordinate of points in C_k, and x_{center} is the horizontal center of the image. This post-processing step is also visualized in Figure 4, where the clusters identified by DBSCAN are classified into left and right lanes, highlighted in green and red, respectively.

3.1.3 Lane Model Fitting

Lane model fitting is critical for accurately modeling lane markings and ensuring smooth lane-keeping control in simulation environments. Since DBSCAN clustering produces a set of discrete lane edge points, these raw data points alone are insufficient for direct use in vehicle control algorithms. By applying mathematical fitting techniques, such as linear regression, this process extracts key parameters, including slope and intercept, providing a continuous and structured representation of lane markings.

In this research, each detected lane cluster undergoes model fitting. Since applying the ROI removes distant regions, the remaining lane markings in the bird's-eye view are often relatively short. Even on curved roads, these short segments can be effectively approximated as straight lines. Therefore, a first-order linear model is preferred for lane representation:

$$ax + by + c = 0 \quad (18)$$

where m is the slope, and c is the intercept. To estimate these parameters, least squares regression is applied by minimizing the sum of squared errors between the observed lane points and the fitted line:

$$\min_{a,b,c} \sum_i \left(\frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}} \right)^2 \tag{19}$$

where (x_i, y_i) are lane pixel coordinates.

The output of this lane perception module includes fitted lane coefficients, which define the slope and intercept of each detected lane boundary. Additionally, it provides clustered lane points for visualization, enabling further analysis and validation of the lane detection process.

3.2 Feedback-based Loop Controller for Lane Centering

The control module is responsible for generating low-level steering and throttle/brake commands to maintain lane centering and achieve the target speed. It consists of two controllers: a lateral controller, which utilizes detected lane boundaries from the perception module to regulate steering, and a longitudinal controller, which modulates vehicle speed by adjusting throttle and braking. Since the vehicle must be fully controlled to operate autonomously within the CARLA simulation environment, implementing both controllers is essential.

We adopt a feedback-based control loop, as illustrated in Figure 5, where the control system operates in a closed-loop configuration. The system first computes the control error $e(t)$, defined as the difference between the desired state $r(t)$ and the measured state $y(t)$. This error signal is then processed by the controller, which consists of proportional, integral, and derivative components. The outputs of these components are summed to generate the control command $u(t)$, which is sent to the process (e.g., vehicle steering actuator) to execute the corrective action. The system response is continuously measured and fed back into the loop, dynamically updating the error for subsequent control adjustments. This feedback mechanism follows the proportional–integral–derivative controller architecture, enabling the controller to dynamically adjust system responses and maintain stable lane-keeping and speed control under varying conditions.

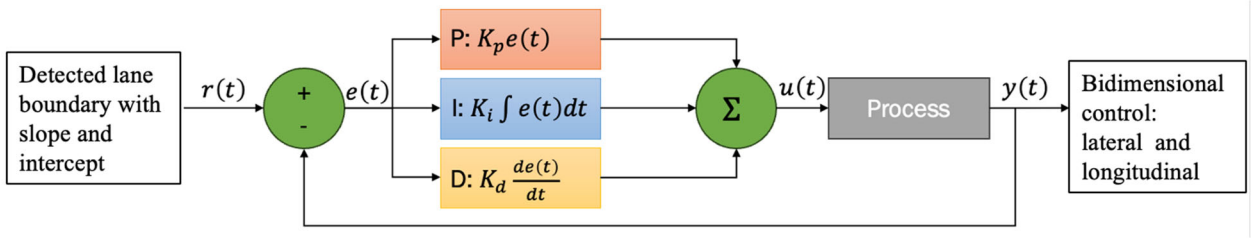


Figure 5. Illustration of feedback-based control loop

The control system is implemented within the CARLA simulation environment, utilizing the vehicle's real-time state to iteratively adjust steering and acceleration for precise lane-keeping.

The feedback-based control mechanism continuously adjusts the control input to minimize the error between the desired and actual values. The control law is given by:

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt} \tag{20}$$

where $e(t)$ represents control error. This equation consists of three terms that contribute to the control response. The proportional term $(K_p e(t))$ provides an immediate corrective action based on the magnitude of the error. A higher K_p results in a stronger response but may lead to oscillations if too large. The integral term $(K_i \int e(t)dt)$ accumulates the sum of past errors and corrects steady-state deviations. However, excessive integral action can lead to overshooting and slow response. The derivative term $(K_d \frac{de(t)}{dt})$ anticipates future errors by considering the rate of change of the error, helping to dampen oscillations. A higher K_d helps prevent overshooting but can amplify noise if not tuned properly. To ensure optimal control performance, the parameters must be fine-tuned to balance responsiveness, stability, and robustness.

Building on this control mechanism design, we further develop two distinct controllers. The first controls lateral movement by adjusting steering to minimize lane deviation, while the second focuses on the longitudinal dimension, regulating throttle and braking to maintain the target speed.

Firstly, the lateral controller determines the required steering angle by considering both the vehicle's lateral deviation from the lane center and its orientation relative to the lane direction.

Before passing the detected lane features from the perception module to the controller, a validation step is performed to ensure the consistency and reliability of lane detection. Since lane detection is susceptible to noise and temporary occlusions, newly detected lanes are compared against previous frame lane positions. If the change in lane orientation exceeds a predefined threshold, the detection is considered unreliable and is smoothed to prevent false inputs from affecting the control process.

The lane center position, denoted as x_{center}, is computed based on the detected left and right lane boundaries. When both lane boundaries are detected, the lane center is given by:

$$x_{center} = \frac{x_{left} + x_{right}}{2} \tag{21}$$

where x_{left} and x_{right} represent the x-coordinates of the detected left and right lane boundaries at the reference points, respectively.

If only one lane boundary is detected, the lane center is estimated using a fixed lane width assumption:

$$x_{center} = \begin{cases} x_{left} + \frac{W_{lane}^{pixel}}{2}, if\ only\ left\ lane\ detected \\ x_{right} - \frac{W_{lane}^{pixel}}{2}, if only\ right\ lane\ detected \end{cases} \tag{22}$$

where W_{lane}^{pixel} represents the known lane width in pixels.

Similarly, the lane direction, denoted as θ_{lane}, is estimated using the slopes of the detected lane boundaries. When both lane boundaries are available, the average of their slopes provides a robust estimate of lane orientation:

$$\theta_{lane} = \frac{\theta_{left} + \theta_{right}}{2} \tag{23}$$

where θ_{left} and θ_{right} are the slopes of the left and right lane boundaries.

If only one lane boundary is detected, the lane direction is inferred from the available boundary:

$$\theta_{lane} = \begin{cases} \theta_{left}, if\ only\ left\ lane\ detected \\ \theta_{right}, if only\ right\ lane\ detected \end{cases} \tag{24}$$

If neither lane boundary is detected, the system defaults to maintaining the previous control command to prevent erratic steering corrections.

The lateral deviation is computed as the difference between the lane center and the vehicle's position:

$$e_{lateral} = x_{center} - x_{vehicle} \tag{25}$$

Since lane detection occurs in pixel coordinates, lateral deviation must be converted to real-world units using the lane width in meters and pixels:

$$e_{lateral}^{meter} = e_{lateral} \times \left(\frac{W_{lane}^{meter}}{W_{lane}^{pixel}} \right) \tag{26}$$

The angle deviation is computed as:

$$e_{angle} = \theta_{lane} - \theta_{vehicle} \tag{27}$$

Since the camera is mounted in the middle of the vehicle, $x_{vehicle}$ is the center of the image and $\theta_{vehicle}$ is fixed at $\frac{\pi}{2}$ radians in bird's-eye view.

The final lateral error used by the controller is a combination of both the lateral deviation and angle deviation:

$$e_{combined} = \gamma e_{lateral}^{meter} + \delta e_{angle} \tag{28}$$

The coefficient γ ensures that lateral deviation correction is appropriately scaled, while δ adjusts the correction intensity based on the vehicle's heading deviation. These coefficients are tuned to balance responsiveness and stability. Using both lateral and angular deviations is essential for effective vehicle control. If the controller only corrects lateral deviation, the vehicle may oscillate due to heading misalignment. Conversely, correcting only angular deviation could cause the vehicle to drift away from the lane center. The combination of these errors provides a more comprehensive measure of vehicle deviation from its desired trajectory.

To maintain control stability, the final steering command is regulated to prevent abrupt changes and excessive steering adjustments. A rate limiter is applied to constrain the steering variation between consecutive frames:

$$\begin{cases} if\ u_{lateral} > u_{prev} + 0.1, u_{lateral} = u_{prev} + 0.1 \\ if\ u_{lateral} < u_{prev} - 0.1, u_{lateral} = u_{prev} - 0.1 \end{cases} \tag{29}$$

where u_{prev} is the previous steering command. Finally, the steering output is clipped to remain within the allowed steering limits:

$$\begin{cases} if\ u_{lateral} \geq 0, steering = \min(\max steer\,, u_{lateral}) \\ if\ u_{lateral} < 0, steering = \max(-\max steer\,, u_{lateral}) \end{cases} \tag{30}$$

These constraints ensure smooth steering transitions and prevent excessive corrections, leading to improved stability and control robustness.

Moving on to the longitudinal controller. It ensures speed tracking by controlling vehicle acceleration and braking. The control error is defined as the difference between the target speedv_{target} and the current speed $v_{current}$:

$$e_{speed} = v_{target} - v_{current} \tag{31}$$

This control error is passed to the longitudinal controller, which computes the required corrective action to regulate speed. To prevent abrupt changes in vehicle dynamics, the computed control output $u_{longitudinal}$ is constrained within valid throttle and brake limits:

$$command = \begin{cases} \min(u_{longitudinal}, \max throttle)\,, if\ u_{longitudinal} \geq 0 \\ \min\left(-u_{longitudinal}, \max brake\right), if\ u_{longitudinal} < 0 \end{cases} \tag{32}$$

This formulation ensures that acceleration remains within allowable throttle limits, while braking is restricted to the maximum brake force when negative.

3.3 Ensemble Learning-based Performance Evaluation Module

The evaluation module assesses LKAS performance under varying environmental and operational conditions, including road geometry, weather, vehicle speed, and lighting. By analyzing vehicle behavior, it quantifies system robustness and limitations, providing insights into LKAS effectiveness. The CARLA simulation environment enables precise customization of these factors, facilitating a controlled and repeatable testing process. To systematically model and interpret the impact of these factors, this research employs Random Forest Regression for performance prediction and SHapley Additive exPlanations (SHAP) analysis to explain individual feature contributions.

3.3.1 Random Forest Regression

To capture complex relationships between input conditions and LKAS performance, we adopt an ensemble-based regression approach using a Random Forest model (Breiman, 2001). This method constructs multiple decision trees, each trained on different subsets of data, to improve predictive stability and generalization while mitigating overfitting.

A decision tree recursively partitions the dataset into subsets by selecting optimal feature splits that minimize prediction error. At each split, the feature value that maximizes variance reduction is selected, formulated as:

$$\Delta V = V_{parent} - \left(\frac{N_{left}}{N} V_{left} + \frac{N_{right}}{N} V_{right}\right) \tag{33}$$

where V represents the variance of the target variable, N is the number of samples in the parent node, and N_{left} and N_{right} are the number of samples in the left and right child nodes, respectively. The tree continues to grow until a stopping criterion is met, such as reaching a minimum number of samples per leaf or a maximum tree depth.

The tree growth continues until a predefined stopping criterion is met, such as a minimum number of samples per leaf or maximum tree depth.

To enhance model robustness, we apply bagging (bootstrap aggregation), where multiple training sets are generated by sampling with replacement from the original dataset. Additionally, feature randomness is introduced by selecting a random subset of features at each split, reducing tree correlation and improving generalization. The final prediction is obtained by averaging predictions from all individual trees:

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} f_t(X) \tag{34}$$

where T is the number of trees, and $f_t(X)$ is the prediction from tree t.

Beyond prediction, feature importance analysis is incorporated to determine which environmental and operational factors most influence LKAS performance. The importance score of a feature x_j is derived by evaluating its contribution to variance reduction across all splits:

$$I_j = \sum_{t=1}^{T} \sum_{s \in S_t} \frac{N_s}{N} \Delta V_s \tag{35}$$

where S_t represents all splits in tree t that use feature x_j, N_s is the number of samples affected by split s, and ΔV_s is the reduction in variance at that split. This metric is known as the Random Forest Mean Decrease in Impurity (MDI) importance. Features with higher I_j values exert greater influence on the model's predictions.

3.3.2 SHAP Analysis

While Random Forest regression provides high predictive accuracy and Built-in MDI feature importance (Eq. 35), it does not explicitly explain how individual input features influence predictions. To address this, SHAP analysis (Lundberg & Lee, 2017) is applied as a post-hoc interpretability method, offering detailed insights into feature contributions.

SHAP is based on cooperative game theory, where the contribution of each feature is quantified using Shapley values, originally formulated to distribute rewards fairly among participants in a game. In our context, the "game" is the prediction process, and the "players" are the input features. The SHAP value of a feature x_j in instance i is computed as:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} \left(f(S \cup \{j\}) - f(S)\right) \tag{36}$$

where F is the set of all features, S is a subset of features excluding x_j, and $f(S)$ represents the model output when only features in S are included. The formula ensures that each feature's contribution is weighted based on all possible feature subsets, providing a fair allocation of importance.

SHAP analysis offers two key interpretability perspectives. First, it provides local explanations by illustrating how each feature contributes to individual predictions, helping to understand the role of specific environmental and operational conditions in determining LKAS performance. Second, it enables global feature importance analysis by computing the mean absolute SHAP value (mean $|\phi_j|$) across all samples, which quantifies the overall impact of each feature on the model output.

By integrating Random Forest regression with SHAP analysis, this research establishes a comprehensive evaluation framework for assessing LKAS performance. Random Forest ensures accurate performance modeling by capturing nonlinear interactions between environmental and operational conditions. Meanwhile, SHAP analysis enhances interpretability by explaining how individual factors contribute to vehicle behavior, offering a detailed understanding of the relationship between input conditions and system response. This combined approach enables a data-driven assessment of LKAS effectiveness, delivering valuable insights for optimizing system performance, improving robustness, and enhancing safety validation.

4. EXPERIMENTAL SETUP AND GENERATION OF DIVERSE SCENARIOS

The performance of LKAS is influenced by multiple environmental and operational factors, among which lane marking characteristics have been extensively studied in previous research. However, real-world testing is inherently constrained by the unpredictability and rarity of certain extreme conditions. In this study, we leverage the highly configurable CARLA simulation environment to systematically generate a diverse set of scenarios, allowing for precise control over weather conditions, road geometry, vehicle speed, streetlights, and vehicle lighting. This capability ensures a comprehensive and repeatable evaluation of LKAS performance across a broad spectrum of driving conditions.

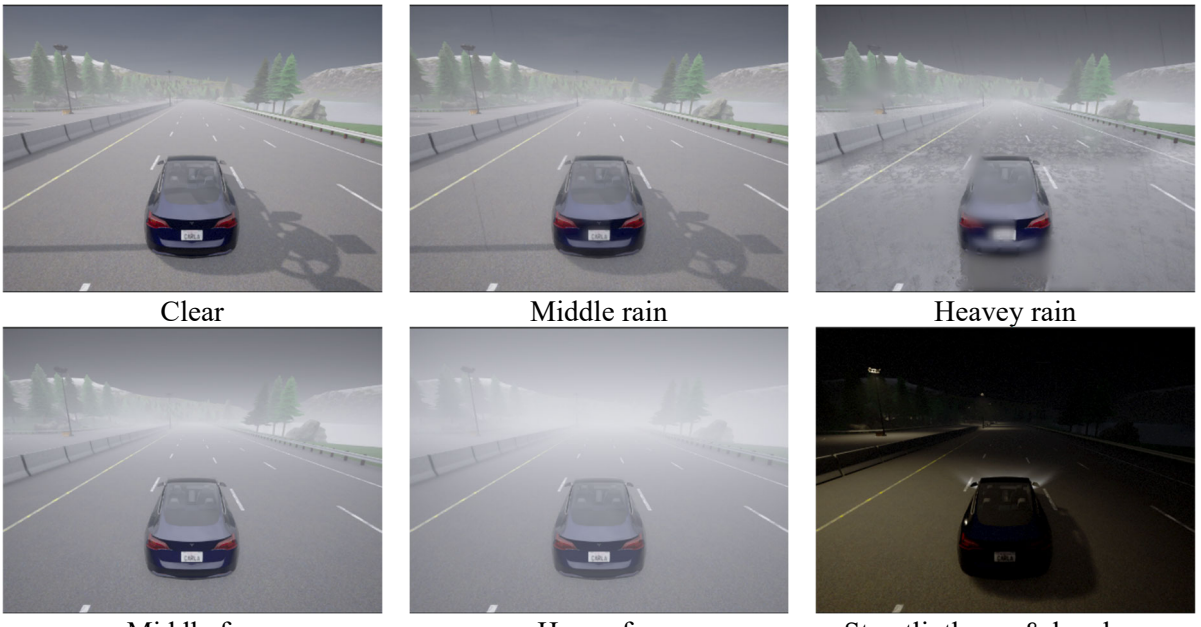
To maximize scenario diversity, experiments are conducted in two distinct CARLA maps: Town03, representing an urban environment, and Town04, simulating a highway setting (See Figure 7). Beyond differences in speed limits (60 km/h vs. 90 km/h), the two maps also differ in road geometry (e.g., four-lane undivided vs. eight-lane divided), lighting placement (roadside vs. median), and roadside scenery (urban vs. open highway), enhancing the realism and variety of driving conditions. Within each map, 36 unique scenarios are systematically generated by varying key driving conditions, capturing a wide range of realistic and challenging driving contexts. Each scenario is executed three times to ensure statistical reliability, resulting in a total of 108 experimental runs per map. By examining LKAS behavior across these controlled yet diverse conditions, this study provides valuable insights into system performance, robustness, and areas for potential improvement.

CARLA offers fine-grained control over weather conditions, enabling realistic simulation of adverse driving environments that would be difficult to test systematically in real-world settings. The weather configurations in this study include clear weather, moderate rain, heavy rain, moderate fog, and heavy fog, each designed to introduce varying degrees of visibility reduction and road surface changes. These weather conditions are simulated by modifying atmospheric and lighting parameters, such as cloud cover, precipitation intensity, fog density, and wetness levels, as summarized in Table 1. These configurations are based on predefined weather presets provided in the CARLA documentation. Each preset consists of a specific combination of parameter values to represent plausible and repeatable environmental scenarios for simulation-based testing.

Table 1. Overview of Weather Parameters in CARLA

Parameters	Data Type	Explanation
Cloudiness	Float (0-100)	Controls the intensity of the clouds
Precipitation	Float (0-100)	Controls the strength of the precipitation
Precipitation deposits	Float (0-100)	Determines the creation of puddles
Wind intensity	Float (0-100)	Controls the strength of the wind
Sun azimuth angle	Float (0-360)°	The azimuth angle of the sun
Sun altitude angle	Float (-90-90)°	Altitude angle of the sun
Fog density	Float (0-100)	Fog concentration or thickness
Fog distance	Float (meters)	Fog starts distance
Wetness	Float (0-100)	Wetness intensity
Fog falloff	Float	Density of the fog
Scattering intensity	Float	Controls how much the light will contribute to volumetric fog
Mie scattering scale	Float	Controls interaction of light with large particles
Rayleigh scattering scale	Float	Controls interaction of light with small particles
Dust storm	Float (0-100)	Determines the strength of the dust storm weather

Each weather condition is carefully tuned using the CARLA API to ensure realistic variations, as illustrated in Figure 6. The clear weather scenario represents an optimal driving environment with minimal atmospheric interference, providing a baseline for LKAS performance evaluation. Moderate rain introduces reduced contrast in lane markings due to surface reflections, while heavy rain significantly affects lane visibility by increasing surface wetness and water pooling. Moderate fog simulates a partially obstructed driving environment, posing challenges for camera-based lane detection as lane boundaries fade into the fog. Heavy fog further amplifies these difficulties, drastically reducing the visible range and increasing LKAS reliance on sensor-based estimation.



Clear | Middle rain | Heavey rain

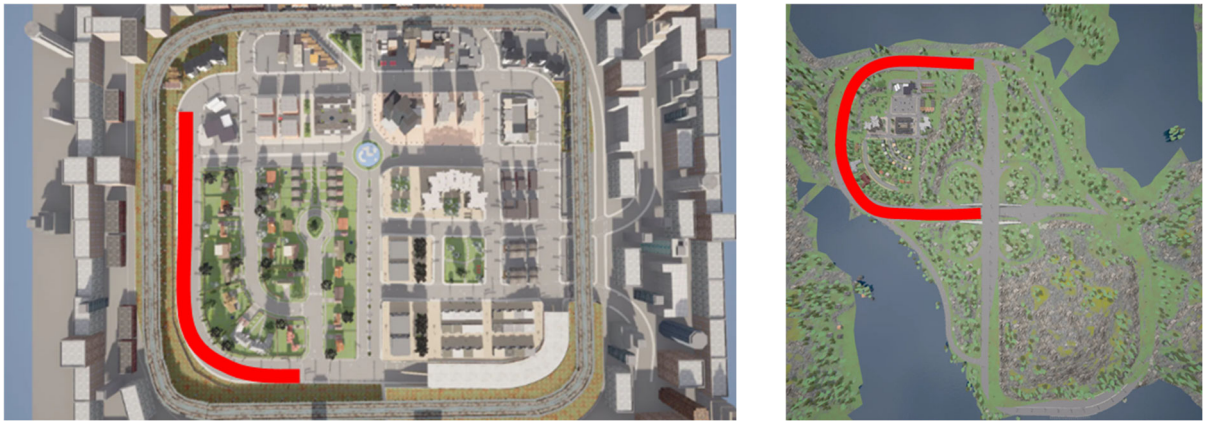
Middle fog | Heavy fog | Streetligths on & low beam



Streetligths on & high beam Streetligths off & low beam Streetligths off & high beam

Figure 6. Overview of Weather Conditions

In addition to weather variations, the experimental setup incorporates road geometry diversity by including both straight and curved segments in each route. Curved roads present a greater challenge for LKAS due to increased lateral acceleration requirements and the reduced visibility of lane boundaries at certain angles. Vehicle speed is also varied to analyze LKAS stability across different driving conditions. In Town03 (urban environment), test speeds are set to 30, 40, 50, and 60 km/h, while in Town04 (highway setting), a broader range of 30, 50, 70, and 90 km/h is tested. This range allows for a realistic assessment of LKAS performance under different speed regimes, reflecting typical urban and highway driving conditions.



Town03 Town04

Figure 7. Overview of Experimental Maps

Artificial lighting conditions are another key factor examined in this study. Streetlight configurations are varied between streetlights on and streetlights off to assess how external illumination affects LKAS, particularly in nighttime scenarios. Additionally, vehicle lighting settings are adjusted by switching between high beams and low beams, allowing for an evaluation of how different illumination intensities impact lane detection. High beams offer better illumination but may introduce glare artifacts, whereas low beams provide limited visibility, especially under adverse weather conditions. To further diversify test conditions, time of day is incorporated as a variable, with experiments conducted under both daytime and nighttime settings to account for natural lighting variations.

The flexibility of the CARLA simulation environment enables the generation of highly diverse and controlled testing conditions, ensuring a systematic and repeatable evaluation of LKAS performance under a wide array of real-world scenarios. This structured framework not only allows for an in-depth analysis of LKAS limitations but also provides a scalable testing platform for future system enhancements.

5. RESULTS AND DISCUSSIONS

5.1 Performance Validation

To validate the performance of the proposed LKAS, we first conduct experiments under clear weather conditions. The primary evaluation metric is lateral deviation, which measures how well the vehicle remains

centered within the lane. Lower lateral deviation values indicate better lane-keeping performance. Figures 8 and 9 depict the distribution of lateral deviations across different speeds for Town03 (urban scenario, left) and Town04 (highway scenario, right), respectively. In these figures, the x-axis represents lateral deviation (in meters), while the y-axis indicates the percentage of recorded deviations within each range.

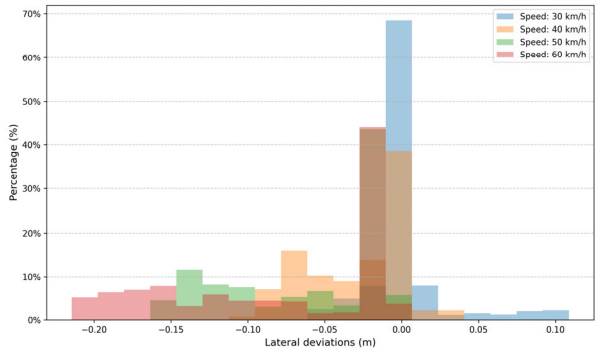


Figure 8. Town03 Lateral Deviation

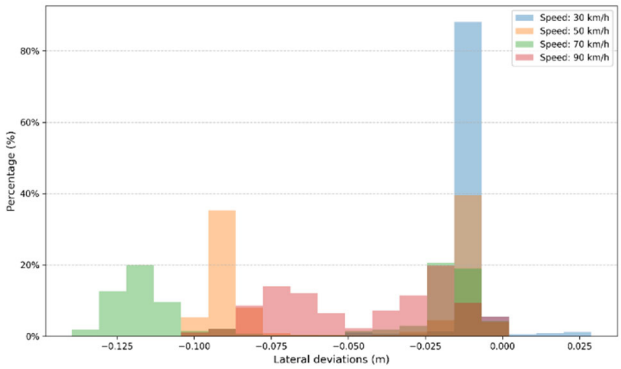


Figure 9. Town04 Lateral Deviation

The results demonstrate that the proposed LKAS maintains strong lane-keeping performance in clear weather, particularly at low to moderate speeds. However, at higher speeds, deviation variations increase, reflecting the greater challenge in maintaining precise lane centering. The system performs slightly better on highway roads (Town04) compared to urban roads (Town03), likely due to the simpler road geometry and fewer external disturbances in highway environments.

Additionally, deviations are predominantly negative across all speed levels, a pattern attributed to the experimental setup. Since all test routes involve left turns without right turns, the deviation distribution is skewed toward negative values, which correspond to the left of the lane centerline. This highlights the influence of route geometry on deviation trends and underscores the need for a balanced route design in future evaluations.

Validations are conducted in two aspects to assess the effectiveness of the proposed LKAS. First, we compared the lateral deviation results with existing studies to ensure consistency with prior findings. The observed lateral deviation values closely align with those reported in the literature. For example, in a separate study, Qi (2024) reported that the lateral deviation of automated vehicles generally ranged between -0.1 and 0.1 m, which is consistent with our results. Similarly, in our earlier research, Chen et al. (2023) found that the lateral wandering distribution of connected and automated vehicles was concentrated between 3 and 4 ft, with over 70% of autonomous vehicles exhibiting lateral deviations of less than 0.3 ft. This also aligns with the outputs form the proposed SimLKAS framework and supports the validity of our findings.

To further evaluate the effectiveness of the proposed LKAS, we compared its lateral deviation performance against that of human-driven vehicles (HDVs). The HDV trajectory data was extracted from the Next Generation SIMulation (NGSIM) dataset, a widely used real-world dataset containing vehicle trajectories collected from major highways such as US-101 and I-80. As shown in Figure 10, the lateral deviation distribution of HDVs follows a normal-like shape, with most values concentrated around 0.0 m but extending up to ±1.5 m. This variability reflects the natural inconsistencies in human driving behavior, influenced by steering corrections, reaction time variations, and external distractions. In contrast, the proposed LKAS exhibits significantly more constrained lane-centering performance, maintaining a tighter deviation range, even at higher speeds. This indicates that LKAS provides superior lane-centering accuracy compared to human drivers in controlled conditions, further reinforcing its potential benefits in reducing unintended lane departures.

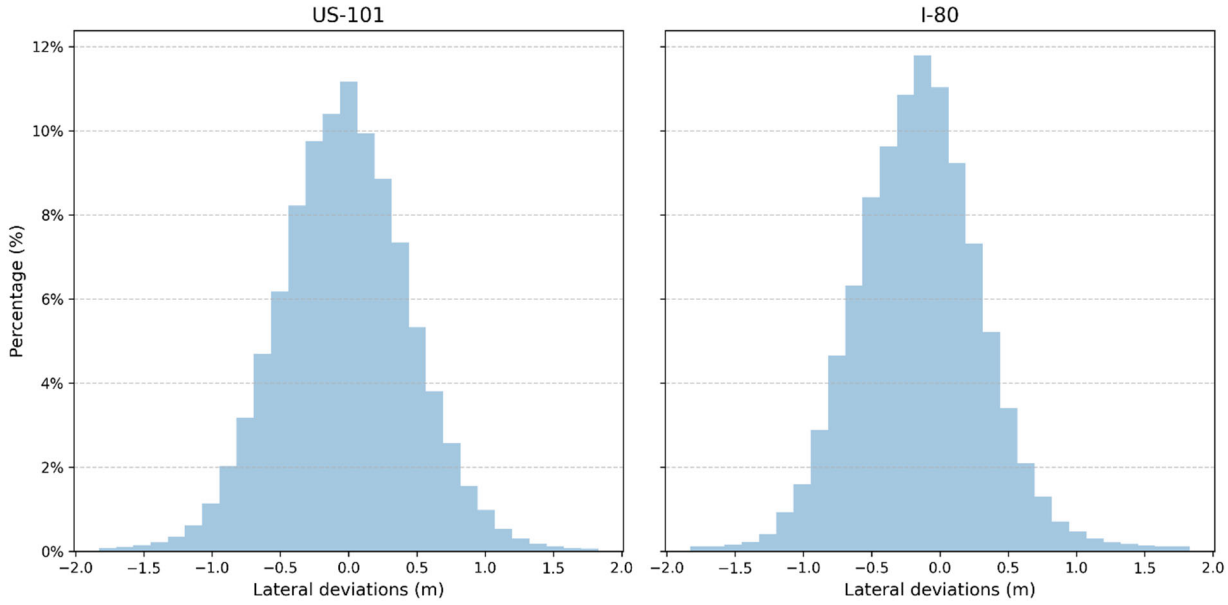


Figure 10. Lateral Deviation Distribution for HDVs on US-101 and I-80

5.2 Performance Under Different Conditions

To assess the impact of weather conditions, road geometries, vehicle speeds, and lighting conditions on the proposed LKAS, we analyze its performance in terms of lateral deviation, where lower deviations indicate better lane-keeping performance. The lateral deviation distributions under these conditions are presented in Figures 11 and 12. Baseline performance is established under clear weather conditions, with all other scenarios analyzed in comparison to this benchmark.

In clear weather, lateral deviations remain minimal across all speeds, particularly on straight roads, where the deviation is consistently close to 0.0 m. However, on curved roads, deviations increase significantly, especially at higher speeds, due to the greater steering corrections required to maintain lane centering.

1) Impact of rain conditions

Under moderate rain, lateral deviation trends remain similar to those observed in clear weather, although higher speeds introduce greater variance, particularly on curved roads. In heavy rain, deviations become more pronounced across both straight and curved roads, with high-speed scenarios exhibiting larger deviations, suggesting reduced LKAS effectiveness due to diminished lane marking visibility. Notably, in Town04, under heavy rain conditions, fewer than 20% of the total route length is completed in certain cases due to severe weather effects, leading to a significant reduction in data points—particularly for curved road segments. This introduces potential performance variability, as seen in the Hard Rain sub-figures of Figure 12.

2) Impact of fog conditions

In light fog, LKAS performance remains comparable to clear weather conditions, indicating stability under moderate visibility reductions. However, heavy fog introduces greater deviation variability, particularly on curved roads, though its overall impact on LKAS performance is less severe than that of heavy rain. Unlike rain-induced reflections and occlusions, fog primarily affects long-range visibility, while LKAS relies more on detecting lane markings at shorter distances. Consequently, all simulations in foggy conditions are successfully completed, and the occurrence of large deviations is less frequent compared to heavy rain conditions.

3) Impact of nighttime lighting conditions

Under nighttime conditions, the absence of streetlights leads to increased lateral deviation, particularly at higher speeds and on curved roads. Low beam headlights provide sufficient illumination on straight roads, but lane deviation increases on curved segments, indicating inadequate lighting for precise lane detection in turns. With high beam headlights, LKAS performance significantly improves, resulting in lateral deviations comparable to those in clear daytime conditions. This underscores the importance of adequate lighting in maintaining lane-tracking accuracy during nighttime driving.

In summary, adverse weather conditions, such as rain and fog, negatively affect LKAS performance, with higher speeds and curved roads exacerbating lateral deviation. Heavy rain poses a greater challenge than heavy fog, as rain affects lane visibility across all distances, while fog primarily impairs long-range vision. Additionally, water accumulation covering lane markings in heavy rain further degrades performance. Lower speeds improve lane-keeping stability, particularly under foggy and nighttime conditions. Sufficient lighting, including high beams and streetlights, significantly enhances LKAS performance at night by improving lane visibility. Overall, while LKAS remains effective in moderate weather conditions, further optimization is needed for extreme scenarios, particularly heavy rain, to enhance lane detection reliability and stability.

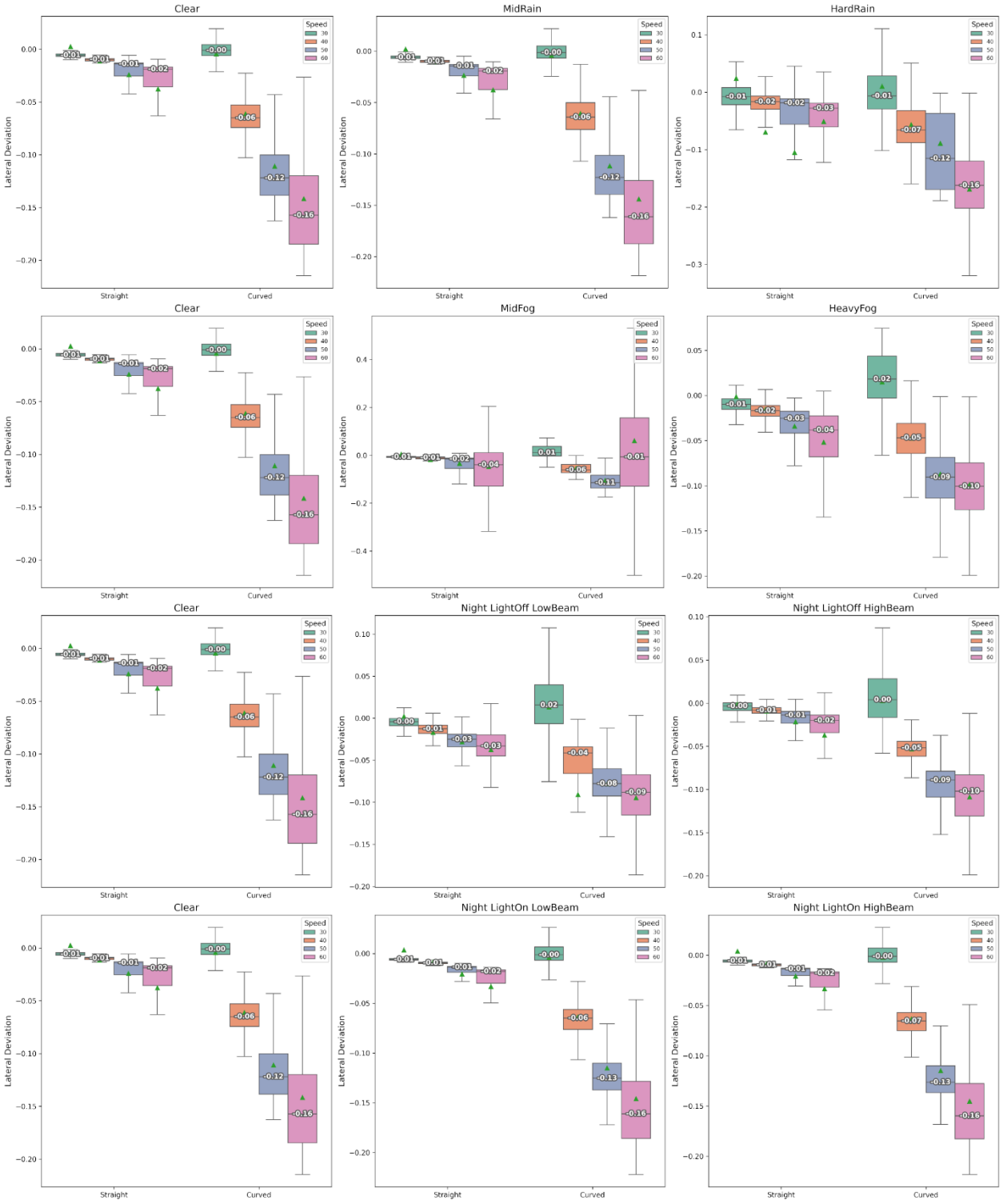


Figure 11. Grouped Boxplots of LKAS Performance in Town03

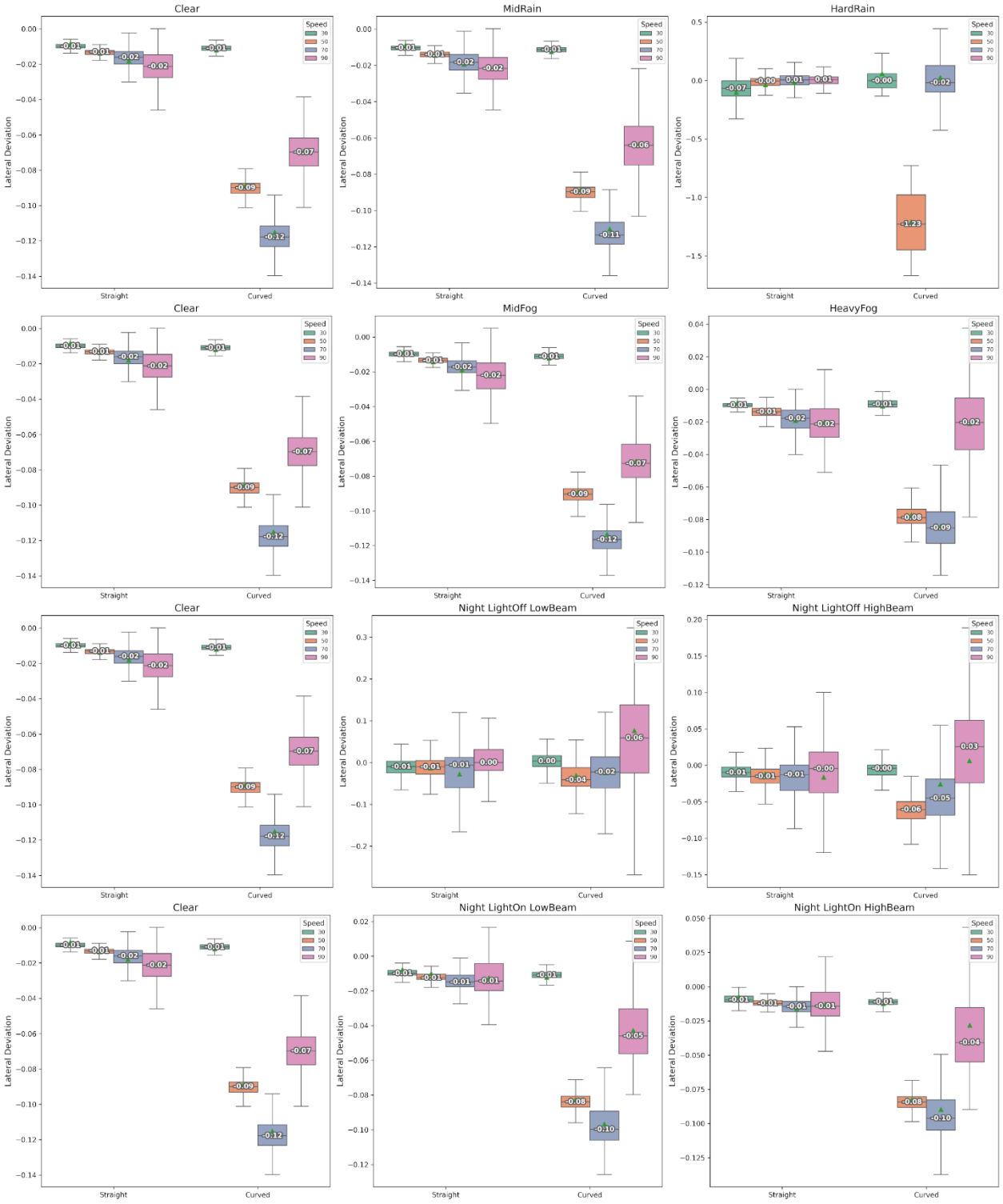


Figure 12. Grouped Boxplots of LKAS Performance in Town04

5.3 Performance Impact Analysis

To understand which environmental and operational factors most strongly affect LKAS performance, a Random Forest model is used in combination with SHAP analysis. Feature importance is evaluated using two complementary approaches: the built-in Random Forest MDI (Eq. 35) and the mean absolute SHAP value $|\phi_j|$ (Eq. 36).

The dependent variable is the maximum lateral deviation, while the independent variables include speed, weather, road geometry, streetlight, and vehicle light conditions. Two separate models are built: one for daytime conditions and one for nighttime conditions. The model performance is evaluated using R^2 and mean squared error (MSE). The daytime model achieves an R^2 of 0.63 and an MSE of 0.090, while the nighttime model has a R^2 of 0.31 and an MSE of 0.080.

Figure 13 shows the feature importance derived from the Random Forest MDI method. For the daytime model, the most influential factor is the heavy rain condition, which accounts for 52.4% of the total feature importance in the model. This is followed by *target_speed* at 35.3%, while *road_Straight* contributes 10.5%. Fog and light rain conditions show minimal impact, indicating that LKAS is less affected by these moderate weather changes.

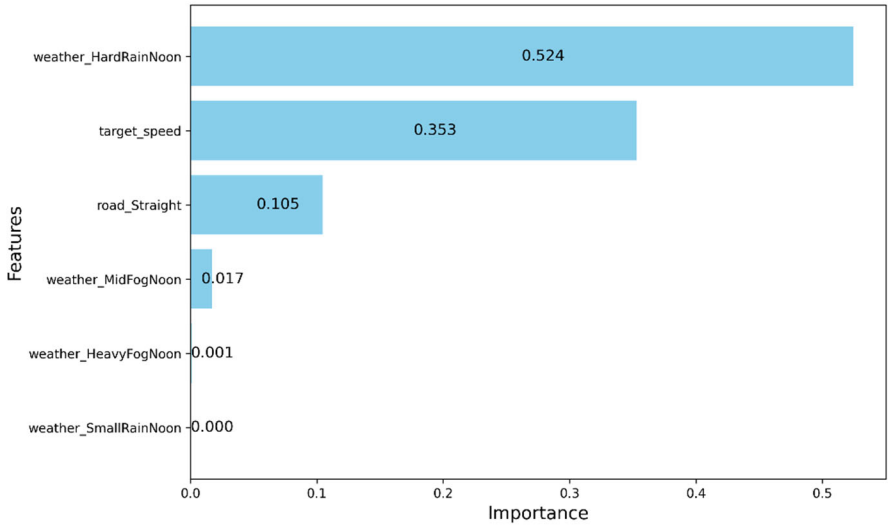


Figure 13. Random Forest MDI Feature Importance for the Daytime Model

Figure 14 presents an SHAP summary plot, where features are ranked by their mean absolute SHAP values. This post-hoc analysis confirms the dominant influence of *weather_HardRainNoon* and *target_speed*, consistent with the MDI results. Unlike MDI, SHAP also reveals directional effects. For categorical variables, high values indicate the presence of the condition. For example, the presence of heavy rain consistently results in positive SHAP values, indicating increased lateral deviation. This reflects the reduced visibility and water accumulation caused by rainfall, which challenge LKAS performance. Similarly, higher vehicle speeds are associated with larger deviations. For *road_Straight*, a high value represents a straight road, while a low value implies a curved one; negative SHAP values for straight roads suggest that curved roads increase lateral deviation, aligning with the statistical findings in section 5.2.

While the top-ranked features are consistent between the MDI (Figure 13) and SHAP (Figure 14) methods, particularly the dominant roles of *weather_HardRainNoon* and *target_speed*, their relative rankings differ beyond the top two. SHAP assigns slightly greater importance to *weather_MidFogNoon*, and slightly less to *road_Straight*, compared to MDI. One possible explanation is that MDI may overestimate the importance of road geometry due to correlations with other variables, such as vehicle speed, which tends to be higher on straight roads. In contrast, SHAP attributes importance more evenly in the presence of correlated features by considering all possible feature combinations.

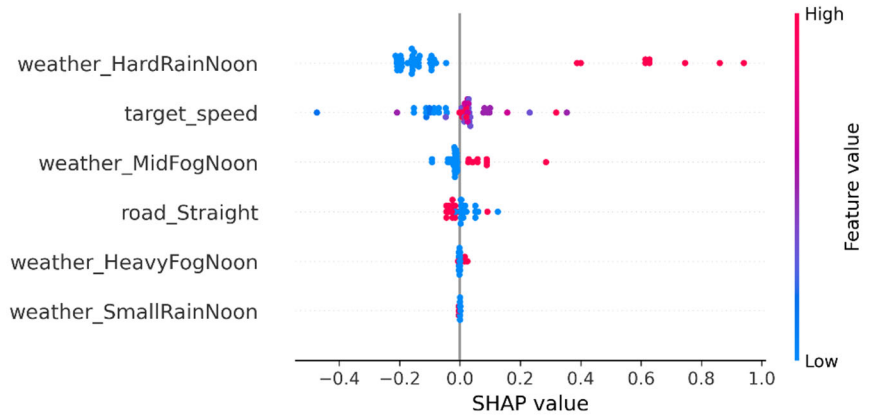


Figure 14. SHAP Summary Plot for the Daytime Model

The MDI feature importance ranking in Figure 15 highlights key contributors to LKAS performance under nighttime conditions. Since weather conditions remain unchanged in nighttime experiments, *target_speed* becomes the most influential factor, contributing for 39.9% of the total feature importance in the nighttime model. Road geometry remains significant, contributing 22.7%. The *streetlight_on* condition is slightly less important than road geometry, contributing 20.9% to the model. Vehicle lighting conditions also ontribute meaningfully, accounting for16.5% of the total importance.

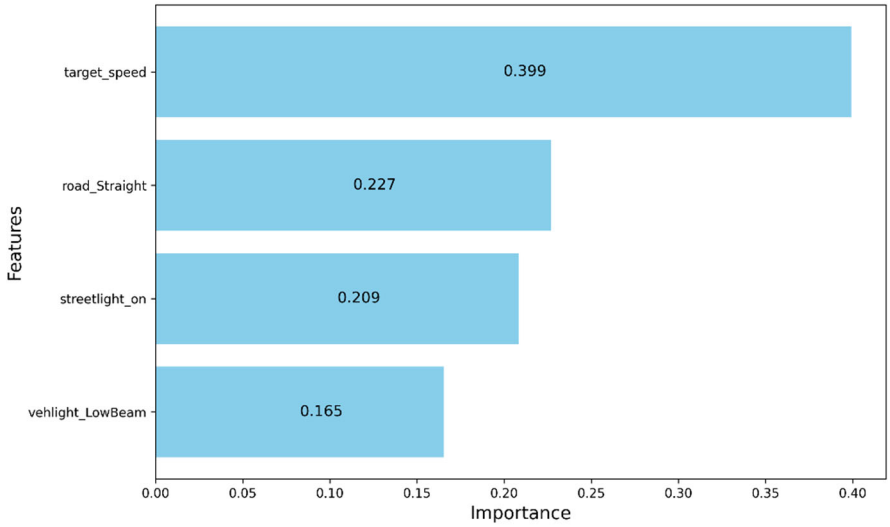


Figure 15. Random Forest MDI Feature Importance for the Nighttime Model

Figure 16 presents an SHAP summary plot for the nighttime model, with features again ranked by their mean absolute SHAP values. For the variable *streetlight_on*, red dots indicate the presence of streetlights (value = 1), while blue dots represent absence (value = 0). The SHAP values reveal that streetlight presence consistently reduces lateral deviation, as indicated by the cluster of negative SHAP values associated with red dots. *Target_speed* also continues to play a significant role. Higher speeds correspond to larger lateral deviations, with SHAP values increasing consistently with higher speed values. The variable *road_Straight* shows a similar pattern as in the daytime model, lower values (curved roads) are associated with positive SHAP values, indicating increased deviation on curves. The separation in SHAP values is more distinct than in the daytime model, suggesting increased sensitivity of the LKAS system to road geometry conditions. The presence of low beam headlights, indicated by high values of *vehlight_LowBeam*, is associated with increased maximum deviations, while using high beams corresponds to improved lane-keeping performance.

Notably, SHAP ranks streetlight presence as the most influential factor, whereas MDI assigns it lower importance. One possibility is that MDI may over-attribute importance to *target_speed* due to its tendency to dominate early splits in the decision trees.

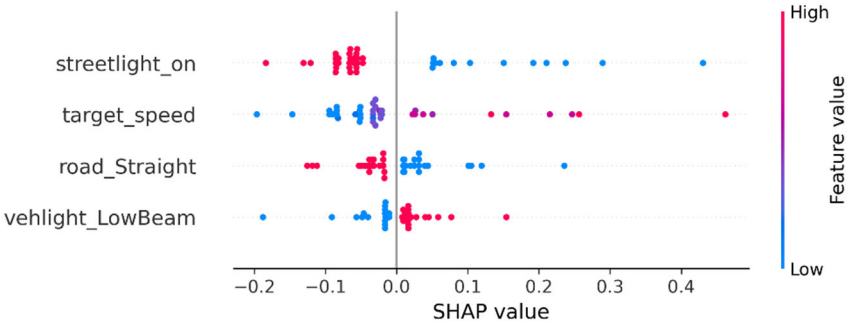


Figure 16. SHAP Summary Plot for the Nighttime Model

The analysis of both models indicates that heavy rain, high speed, and curved roads are the primary factors contributing to increased maximum deviation in daytime conditions. At night, visibility-related factors, such as streetlights and vehicle headlights, become more critical in maintaining LKAS performance. Based on these findings, improvements to LKAS can be achieved in several ways. First, lane detection algorithms can be enhanced. Integrating adaptive contrast enhancement and multi-frame temporal filtering could improve lane detection reliability in adverse weather conditions. In rainy or foggy conditions, deep learning-based lane segmentation models trained on diverse weather datasets can enhance robustness against occluded lane markings. Additionally, sensor fusion combining camera and LiDAR or radar data could improve lane visibility under low-light and extreme weather conditions. Second, control algorithms can be optimized. Dynamic speed adjustment based on real-time weather and road curvature estimation can improve lane-keeping stability. Employing reinforcement learning-based adaptive control strategies could improve LKAS performance by learning from diverse environmental conditions and vehicle dynamics. Finally, external techniques can be implemented to support LKAS performance. For example, adaptive headlight control systems could dynamically adjust illumination based on road geometry and visibility conditions, enhancing lane perception at night. V2I (Vehicle-to-Infrastructure) communication could provide real-time road and weather condition updates to optimize LKAS behavior, particularly in areas with poor visibility. Additionally, high-definition (HD) maps with lane-level accuracy could serve as a backup reference for lane-keeping in situations where lane markings are temporarily obscured. These improvements could significantly enhance LKAS performance under challenging weather and nighttime driving scenarios, ensuring more reliable lane-keeping in real-world conditions.

6. CONCLUSION

This study presents a new simulation-based evaluation framework for analyzing LKAS performance under diverse driving conditions. By addressing the limitations of real-world testing, which is often costly, time-consuming, and constrained by natural environmental occurrences, this framework provides a controllable, repeatable, and cost-efficient alternative for evaluating LKAS. The proposed framework consists of three integrated core modules—perception, control, and evaluation—which work in tandem to simulate real-time lane detection, apply lane-keeping control strategies, and assess performance through quantitative analysis. Implemented in the CARLA simulation environment, the framework enables precise control over critical factors such as weather, road geometry, vehicle speed, and lighting conditions, allowing for systematic performance benchmarking in a scalable and reproducible manner.

A key contribution of this research is the systematic generation of diverse testing scenarios, addressing the challenges of collecting real-world data under rare or extreme conditions. The framework successfully generated 108 experimental runs, covering a range of environmental and operational conditions, including varied weather (clear, rain, fog), lighting (streetlights, headlights), and road geometries (straight and curved

roads). Additionally, a comparative analysis of vehicle behavior patterns with findings from field data and prior studies confirmed that the proposed framework accurately reproduces realistic driving behaviors, further validating its effectiveness for LKAS assessment. The results indicate that LKAS performance is most significantly impacted by heavy rain, high speeds, and curved roads, as these factors introduce greater lane deviation and reduce lane detection reliability. Nighttime conditions further affect performance, with streetlight availability and headlight settings playing a critical role in maintaining lane-keeping stability. A detailed SHAP-based feature importance analysis confirms that rain-induced surface reflections, vehicle speed, and road curvature are dominant factors contributing to lateral deviation during daytime driving, whereas visibility-related conditions such as streetlights and headlights are crucial for nighttime lane tracking.

Despite the effectiveness of LKAS in moderate conditions, the findings highlight several areas for improvement to enhance lane-keeping reliability in extreme environments, with three key areas prioritized for future advancements: integrating sensor fusion and deep learning-based segmentation to improve lane detection under adverse weather conditions—where traditional vision-based methods falter due to reflections, occlusions, or low contrast—while incorporating reinforcement learning or predictive control techniques to enable adaptive real-time adjustments for better vehicle stability in dynamic driving scenarios, and leveraging Vehicle-to-Infrastructure (V2I) communication alongside high-definition (HD) maps to provide additional environmental context, ultimately enhancing LKAS robustness in challenging situations.

Practically, the proposed framework can serve as a pre-validation tool in the LKAS development cycle, helping to identify system limitations before field testing. This approach can reduce development costs, shorten the V&V process, and improve testing safety by enabling the exploration of critical scenarios in a controlled environment. Furthermore, the framework can be integrated with traffic simulators such as SUMO to investigate the broader impact of LKAS-equipped vehicles on traffic safety and efficiency. For example, researchers and developers can simulate mixed-traffic environments under diverse environmental and operational conditions to assess how LKAS behavior influences lane-keeping stability, lateral deviations, and potential conflicts. These insights can guide system design improvements and inform policy decisions for real-world deployment.

While the proposed framework offers a flexible and cost-effective platform for LKAS evaluation, it has several limitations. First, simulation tools like CARLA cannot fully replicate real-world complexity. Conditions such as snow, road debris, worn pavement markings, and gradual environmental changes are not well modeled, and sensor behaviors like noise or calibration drift are simplified. Second, LKAS performance in simulation may appear more stable than in practice, as CARLA idealizes actuation and omits hardware integration effects such as latency or mechanical variability. Third, the framework does not simulate sensor malfunctions, actuator failures, or driver-in-the-loop interactions, factors that are critical for assessing LKAS robustness in real-world settings. These limitations highlight that while the framework is not a substitute for real-world validation, it serves as a powerful tool for pre-validation, controlled scenario testing, and broader impact analysis. It can be used not only to identify performance limitations under diverse conditions but also to explore the effects of LKAS behavior on traffic safety and efficiency in simulated mixed-traffic environments.

REFERENCES

Amini, A., Gilitschenski, I., Phillips, J., Moseyko, J., Banerjee, R., Karaman, S., & Rus, D. (2020). Learning robust control policies for end-to-end autonomous driving from data-driven simulation. *IEEE Robotics and Automation Letters*, *5*(2), 1143-1150.

Assidiq, A. A., Khalifa, O. O., Islam, M. R., & Khan, S. (2008). Real time lane detection for autonomous vehicles. 2008 International Conference on Computer and Communication Engineering, IEEE.

Aung, T., & Zaw, M. H. (2014). Video based lane departure warning system using hough transform. International Conference on Advances in Engineering and Technology, Citeseer.

Breiman, L. (2001). Random forests. *Machine learning*, *45*, 5-32.

Burghardt, T. E., Popp, R., Helmreich, B., Reiter, T., Böhm, G., Pitterle, G., & Artmann, M. (2021). Visibility of various road markings for machine vision. *Case Studies in Construction Materials*, *15*, e00579.

Canny, J. (1986). A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*(6), 679-698.

Chen, C., Song, Y., Wang, Y. D., Hu, X., & Liu, J. (2023). A machine learning-based approach to assess impacts of autonomous vehicles on pavement roughness. *Philosophical Transactions of the Royal Society A*, *381*(2254), 20220176.

Chen, Y., Ji, C., Cai, Y., Yan, T., & Su, B. (2024). Deep Reinforcement Learning in Autonomous Car Path Planning and Control: A Survey. *arXiv preprint arXiv:2404.00340*.

Chen, Z., & Huang, X. (2017, 11-14 June 2017). End-to-end learning for lane keeping of self-driving cars. 2017 IEEE Intelligent Vehicles Symposium (IV),

Cheng, S., Li, L., Chen, X., Wu, J., & Wang, H.-d. (2020). Model-predictive-control-based path tracking controller of autonomous vehicle considering parametric uncertainties and velocity-varying. *IEEE Transactions on Industrial Electronics*, *68*(9), 8698-8707.

Chu, D., Li, H., Zhao, C., & Zhou, T. (2022). Trajectory tracking of autonomous vehicle based on model predictive control with PID feedback. *IEEE Transactions on Intelligent Transportation Systems*, *24*(2), 2239-2250.

Ding, D., Lee, C., & Lee, K.-y. (2013). An adaptive road ROI determination algorithm for lane detection. 2013 IEEE International Conference of IEEE Region 10 (TENCON 2013), IEEE.

Formosa, N., Quddus, M., Man, C. K., Singh, M. K., Morton, C., & Masera, C. B. (2024). Evaluating the Impact of Lane Marking Quality on the Operation of Autonomous Vehicles. *Journal of Transportation Engineering, Part A: Systems*, *150*(1), 04023126. https://doi.org/10.1061/jtepbs.Teeng-7688

Ghafoorian, M., Nugteren, C., Baka, N., Booij, O., & Hofmann, M. (2018). El-gan: Embedding loss driven generative adversarial networks for lane detection. proceedings of the european conference on computer vision (ECCV) Workshops,

Ghanem, S., Kanungo, P., Panda, G., Satapathy, S. C., & Sharma, R. (2023). Lane detection under artificial colored light in tunnels and on highways: an IoT-based framework for smart city infrastructure. *Complex & Intelligent Systems*, *9*(4), 3601-3612.

Hadi, M., & Sinha, P. (2011). Effect of pavement marking retroreflectivity on the performance of vision-based lane departure warning systems. *Journal of Intelligent Transportation Systems*, *15*(1), 42-51.

Hou, Y., Ma, Z., Liu, C., & Loy, C. C. (2019). Learning lightweight lane detection cnns by self attention distillation. Proceedings of the IEEE/CVF international conference on computer vision,

Hsiao, P.-Y., Yeh, C.-W., Huang, S.-S., & Fu, L.-C. (2008). A portable vision-based real-time lane departure warning system: day and night. *IEEE Transactions on Vehicular Technology*, *58*(4), 2089-2094.

Kim, J., & Lee, M. (2014). Robust lane detection based on convolutional neural network and random sample consensus. Neural Information Processing: 21st International Conference, ICONIP 2014, Kuching, Malaysia, November 3-6, 2014. Proceedings, Part I 21, Springer.

Kim, Z. (2008). Robust lane detection and tracking in challenging scenarios. *IEEE Transactions on Intelligent Transportation Systems*, *9*(1), 16-26.

Kuutti, S., Bowden, R., Jin, Y., Barber, P., & Fallah, S. (2021). A Survey of Deep Learning Applications to Autonomous Vehicle Control. *IEEE Transactions on Intelligent Transportation Systems*, *22*(2), 712-733. https://doi.org/10.1109/tits.2019.2962338

Li, D., Zhao, D., Zhang, Q., & Chen, Y. (2019). Reinforcement learning and deep learning based lateral control for autonomous driving [application notes]. *IEEE Computational Intelligence Magazine*, *14*(2), 83-98.

Liu, W., Hua, M., Deng, Z., Meng, Z., Huang, Y., Hu, C., Song, S., Gao, L., Liu, C., Shuai, B., Khajepour, A., Xiong, L., & Xia, X. (2023). A Systematic Survey of Control Techniques and Applications in Connected and Automated Vehicles. *IEEE Internet of Things Journal*, *10*(24), 21892-21916. https://doi.org/10.1109/jiot.2023.3307002

Liu, W., Song, L., Dong, Y., Zhang, X., & Xu, L. (2025). Unified model predictive control method of automated vehicles for lane-changing and lane-keeping maneuvers. *Journal of Intelligent Transportation Systems*, 1-21.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, *30*.

Marino, R., Scalzi, S., & Netto, M. (2011). Nested PID steering control for lane keeping in autonomous vehicles. *Control Engineering Practice*, *19*(12), 1459-1467. https://doi.org/10.1016/j.conengprac.2011.08.005

Marr, J., Benjamin, S., & Zhang, A. (2020). *Implications of Pavement Markings for Machine Vision*. Austroads.

MITRE. (2024). *PARTS: Market Penetration of Advanced Driver Assistance Systems (ADAS)*. https://www.mitre.org/sites/default/files/2024-09/PR-24-2614-PARTS-Market-Penetration-Advanced-Driver-Assistance-Systems-0917.pdf

Mu, C., & Ma, X. (2014). Lane detection based on object segmentation and piecewise fitting. *TELKOMNIKA Indonesian Journal of Electrical Engineering*, *12*(5), 3491-3500.

National Highway Traffic Safety Administration. (2024). *Lane Keeping Assist and Lane Departure Warning System Confirmation Test Procedure*. https://downloads.regulations.gov/NHTSA-2024-0077-0002/attachment_13.pdf

Neumeister, D. M., & Pape, D. B. (2019). *Automated vehicles and adverse weather*. https://rosap.ntl.bts.gov/view/dot/43772

Neven, D., De Brabandere, B., Georgoulis, S., Proesmans, M., & Van Gool, L. (2018). Towards end-to-end lane detection: an instance segmentation approach. 2018 IEEE intelligent vehicles symposium (IV), IEEE.

Niu, J., Lu, J., Xu, M., Lv, P., & Zhao, X. (2016). Robust lane detection using two-stage feature extraction with curve fitting. *Pattern Recognition*, *59*, 225-233.

Pérez-Gil, Ó., Barea, R., López-Guillén, E., Bergasa, L. M., Gómez-Huélamo, C., Gutiérrez, R., & Díaz-Díaz, A. (2022). Deep reinforcement learning based control for Autonomous Vehicles in CARLA. *Multimedia Tools and Applications*, *81*(3), 3553-3576.

Pike, A., Shirinzad, M., Nayak, A., & Rathinam, S. (2024). *Assessing Pavement Markings for Automated Vehicle Readiness* [Tech Report]. https://rosap.ntl.bts.gov/view/dot/77435

Pike, A. M., Barrette, T. P., & Carlson, P. J. (2018). Evaluation of the Effects of Pavement Marking Characteristics on Detectability by ADAS Machine Vision. *National Cooperative Highway Research Program (NCHRP): Washington, DC, USA*.

Qi, H. (2024). Lateral Dimension Modeling and Application. In H. Qi (Ed.), *Stochastic Two-Dimensional Microscopic Traffic Model: Theory and Applications* (pp. 117-153). Springer Nature Singapore. https://doi.org/10.1007/978-981-97-3597-6_6

Qi, H., & Hu, X. (2023). Behavioral investigation of stochastic lateral wandering patterns in mixed traffic flow. *Transportation Research Part C: Emerging Technologies*, *155*. https://doi.org/10.1016/j.trc.2023.104310

Sallab, A. E., Abdou, M., Perot, E., & Yogamani, S. (2016). End-to-end deep reinforcement learning for lane keeping assist. *arXiv preprint arXiv:1612.04340*.

Shaon, M. R., Orlova, E., & Jackson, E. (2024). *Automated Vehicle and Pavement Marking Evaluation in Connecticut* [Tech Report]. https://rosap.ntl.bts.gov/view/dot/76725

Tang, J., Li, S., & Liu, P. (2021). A review of lane detection methods based on deep learning. *Pattern Recognition*, *111*, 107623.

Utriainen, R., Pollanen, M., & Liimatainen, H. (2020). The Safety Potential of Lane Keeping Assistance and Possible Actions to Improve the Potential. *IEEE Transactions on Intelligent Vehicles*, *5*(4), 556-564. https://doi.org/10.1109/tiv.2020.2991962

Wang, C., Wang, L., Lu, Z., Chu, X., Shi, Z., Deng, J., Su, T., Shou, G., & Wen, X. (2023). SRL-TR 2: A Safe Reinforcement Learning Based TRajectory TRacker Framework. *IEEE Transactions on Intelligent Transportation Systems*, *24*(6), 5765-5780.

Wang, P., Li, H., & Chan, C.-Y. (2019). Continuous control for automated lane change behavior based on deep deterministic policy gradient algorithm. 2019 IEEE Intelligent Vehicles Symposium (IV), IEEE.

Wang, Y., Alhuraish, A., Yuan, S., Wang, S., & Zhou, H. (2025). OpenLKA: an open dataset of lane keeping assist from market autonomous vehicles. *arXiv preprint arXiv:2501.03287*.

Wang, Y., Wang, J., Yang, Y., Li, Z., & Zhao, X. (2022). An end-to-end deep reinforcement learning model based on proximal policy optimization algorithm for autonomous driving of off-road vehicle. International Conference on Autonomous Unmanned Systems, Springer.

Wei, S., Pfeffer, P. E., & Edelmann, J. (2023). State of the art: Ongoing research in assessment methods for lane keeping assistance systems. *IEEE Transactions on Intelligent Vehicles*.

Wu, P.-C., Chang, C.-Y., & Lin, C. H. (2014). Lane-mark extraction for automobiles under complex conditions. *Pattern Recognition*, *47*(8), 2756-2767.

Zakaria, N. J., Shapiai, M. I., Ghani, R. A., Yassin, M. N. M., Ibrahim, M. Z., & Wahid, N. (2023). Lane Detection in Autonomous Vehicles: A Systematic Review. *IEEE Access*, *11*, 3729-3765. https://doi.org/10.1109/access.2023.3234442

Zhang, R., Wu, Y., Gou, W., & Chen, J. (2021). RS-Lane: A Robust Lane Detection Method Based on ResNeSt and Self-Attention Distillation for Challenging Traffic Situations. *Journal of Advanced Transportation*, *2021*(1), 7544355.

Zhao, P., Chen, J., Song, Y., Tao, X., Xu, T., & Mei, T. (2012). Design of a control system for an autonomous vehicle based on adaptive-pid. *International Journal of Advanced Robotic Systems*, *9*(2), 44.

Zou, Q., Jiang, H., Dai, Q., Yue, Y., Chen, L., & Wang, Q. (2019). Robust lane detection from continuous driving scenes using deep neural networks. *IEEE Transactions on Vehicular Technology*, *69*(1), 41-54.